ST. MARY'S DSG, KLOOF
PHYSICAL SCIENCES
Cycle test

GRADE 10 **TIME: 50 mins**
September 2020 **MARKS: 53**
Examiner: Mrs K. Bolton **Moderators: Mrs G. Main and Mrs T. Kirsten**

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 5 pages. Please check that your question paper is complete.
2. Answer the questions on your double folio sheet provided and number according to the question paper.

Q = C.m.ΔT

QUESTION 1 – MULTIPLE CHOICE

1.1 Which bonds can be attributed to silicon dioxide's high boiling point?

A Metallic bonds
B Covalent bonds
C Hydrogen bonds
D Ionic bonds

1.2 In which of the following compounds is the bond ***most*** covalent in nature?

A $KC\ell$
B $MgC\ell_2$
C $BeC\ell_2$
D $CaC\ell_2$

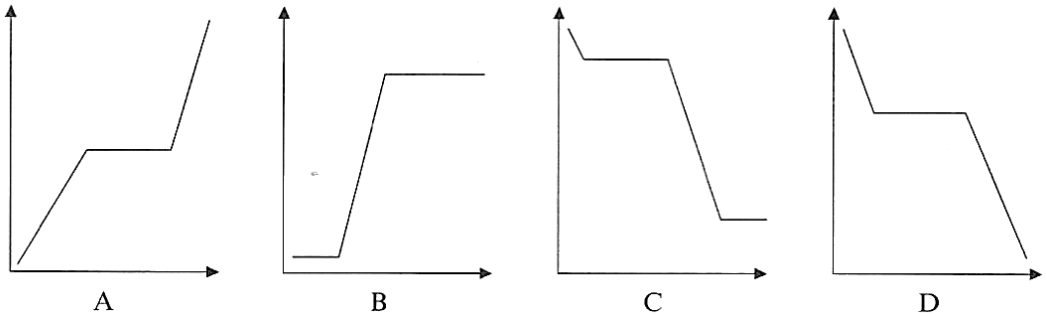
1.3 When you transfer heat to a substance, you always increase its

A energy
B temperature
C specific heat capacity
D both A and B are correct

1.4 Which of the following statements is TRUE?

A A person feels cold when the temperature drops because the cold seeps into their bones.
B A person takes a hot-water bottle to bed so that their body can absorb the heat from it.
C A person feels cold under their arms when deodorant is sprayed on because the deodorant is at a lower temperature than their body temperature.
D A cooldrink becomes much colder when ice blocks are placed in it because the cold from the ice blocks is absorbed by the warmer liquid.

1.5 Which of the following curves is a cooling curve where only one phase change sets in? (y-axis = temperature, x-axis = time)



[2x5=10]

QUESTION 2

The table shows the melting points, boiling points and electrical properties of five substances **A** to **E**.

Substance	Melting point (ºC)	Boiling point (ºC)	Electrical conductivity of solid	Electrical conductivity of liquid
A	-7	59	Poor	Poor
B	1083	2567	Good	Good
C	755	1387	Poor	Good
D	43	181	Poor	Poor
E	1607	2227	Poor	Poor

2.1 Choose a substance from the table above to match each of the following descriptions. A substance may be used once, more than once or not at all. Justify each choice with evidence from the table.

2.1.1 This substance has a giant covalent structure. (2)

2.1.2 This substance is a metal. (2)

2.1.3 This substance is an ionic solid. (2)

2.1.4 This substance has London type intermolecular forces and is a liquid at room temperature (25ºC). (2)

2.2 Water and methane have similar relative molar masses, and both contain the element hydrogen. The table below gives some information about water and methane and carbon dioxide

	H_2O	**CH_4**	**CO_2**
Molar mass ($g.mol^{-1}$)	18	16	44
Melting point (ºC)	0	-182	
Boiling point (ºC)	100	-165,5	-78,5

2.2.1 Draw the Lewis diagram for the water molecule. (2)

2.2.2 Define *intermolecular force*. (2)

2.2.3 State the predominant type of intermolecular force holding the water molecules together in the ice crystal. (1)

2.2.4 State the predominant type of intermolecular force in methane. (1)

2.2.5 Explain why the melting point of ice is higher than the melting point of methane. (2)

2.2.6 Explain why the “boiling point” of carbon dioxide is higher than that of methane. (3)

2.2.7 When dry ice (solid carbon dioxide) is placed in a beaker of water, bubbles are seen and a white cloud. After a few minutes, the temperature of the water in the beaker decreases.



(i) Identify which molecules are responsible for the bubbles in the beaker. (2)

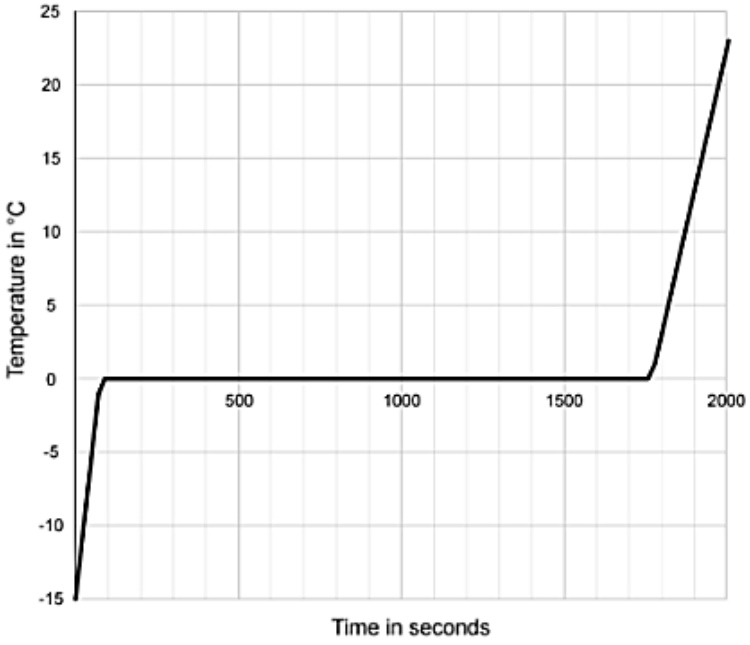
(ii) Explain why the temperature of the water in the beaker decreases after a few minutes. (2)

2.2.8 The figure below shows solid ice on a car's rear window. The glass window contains an electrical heating element.



(i) Define the term *latent heat.* (2)

The graph below shows how the temperature changes as the heater transfers energy to the ice.



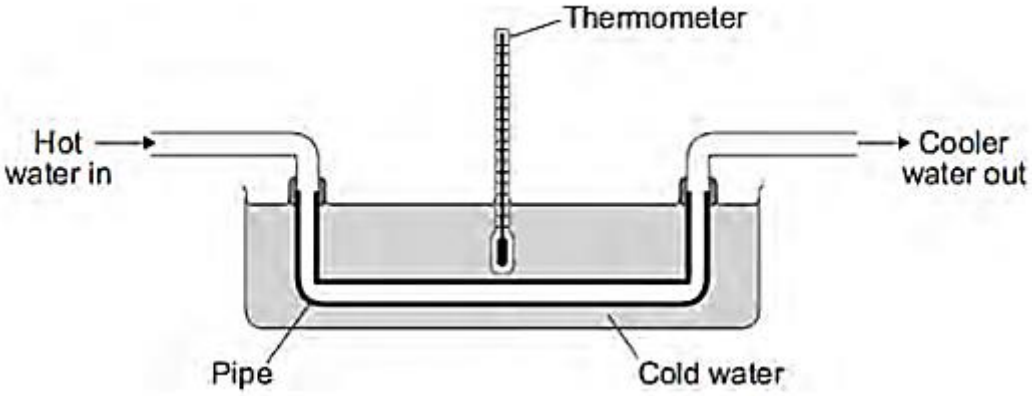
(ii) Between 80 – 1760 seconds the energy supplied by the heater does not cause the temperature of the ice to increase. Explain the reason for this. (2)

(iii) After 1760 seconds the temperature of the water began to increase. Explain the reason for this. (2)

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QUESTION THREE

Heat exchangers are devices used to transfer heat from one place to another. The diagram shows a pipe being used as a simple heat exchanger by a student in an investigation. Heat is transferred from the hot water inside the pipe to the cold water outside the pipe.



The student wanted to find out if the efficiency of a heat exchanger depends on the material used to make the pipe. The student tested three different materials. For each material, the rate of flow of hot water through the pipe was kept the same. The student's results are recorded in the table.

Material	Temperature of the cold water at the start in °C	Temperature of the cold water after 10 minutes in °C
Copper	20	36
Glass	20	23
Plastic	20	21

3.1 Define the term *heat.* (2)

3.2 Write an aim for this investigation. (2)

3.3 What is the independent variable in this investigation? (2)

3.4 Give one controlled variable in the investigation. (1)

3.5 Which one of the three materials made the best heat exchanger? Give a reason for your answer. (2)

3.6 Define the term *specific heat capacity*. (2)

3.7 Using the information for the glass, calculate the amount of heat energy transferred to 500 g of water if the specific heat capacity for water is 4 200 $J.kg^{-1}.°C^{-1}$. (3)

[14]